Monthly Gulf of Mexico OHC/TCHP Comparison
Argo GDAC vs Same-Day RTOFS

Overview

This report documents a TEOS-10-backed collocation study carried out in the Gulf of Mexico for 2024. For each month in January–March and August–October, we selected the top 5 Argo-coverage dates in the month, paired each Argo GDAC profile to the nearest same-day RTOFS ocean column at the profile coordinates, and computed Tropical Cyclone Heat Potential (TCHP) from both profiles using TEOS-10/GSW thermodynamics.

Method

For each profile/column pair:

- Practical Salinity was converted to Absolute Salinity using `gsw_SA_from_SP`.
- Pressure/depth conversion used `gsw_z_from_p` for Argo and `gsw_p_from_z` for RTOFS.
- Density used `gsw_rho_t_exact` and heat capacity used `gsw_cp_t_exact`.
- The depth of the 26 °C isotherm (D_{26}) was found by linear interpolation.
- TCHP was evaluated as $\int_0^{D_{26}} \rho(z) c_p(z) [T(z) - 26]\, dz$ and reported in kJ/cm^2.

Monthly Summary

Month	Pairs	Floats	Dates	Argo mean (kJ/cm^2)	RTOFS mean (kJ/cm^2)	Mean diff (Argo-RTOFS)
Jan	4	4	1	40.13	21.59	18.54
Feb	11	7	5	28.95	20.40	8.54
Mar	18	11	5	26.96	15.77	11.19
Aug	59	19	5	106.94	99.98	6.96
Sep	53	18	5	105.51	95.28	10.24
Oct	42	32	5	80.07	74.51	5.56

Seasonal Means

Window	Pairs	Argo mean (kJ/cm^2)	RTOFS mean (kJ/cm^2)	Mean diff (kJ/cm^2)
Jan-Mar	33	29.22	18.02	11.20
Aug-Oct	154	99.12	91.42	7.70

Selected Sample Dates

Month	Date	Argo profiles
Jan	20240101	14
Jan	20240107	11
Jan	20240111	14
Jan	20240121	14
Jan	20240131	15
Feb	20240201	13
Feb	20240210	16
Feb	20240216	13
Feb	20240220	14
Feb	20240221	14
Mar	20240302	16
Mar	20240307	14
Mar	20240312	16
Mar	20240317	15
Mar	20240322	19
Aug	20240804	13
Aug	20240809	12
Aug	20240814	13
Aug	20240824	12
Aug	20240830	13
Sep	20240904	15
Sep	20240909	13
Sep	20240914	12
Sep	20240920	11
Sep	20240925	11
Oct	20241003	9
Oct	20241004	9
Oct	20241005	10
Oct	20241009	9
Oct	20241016	10

Figures

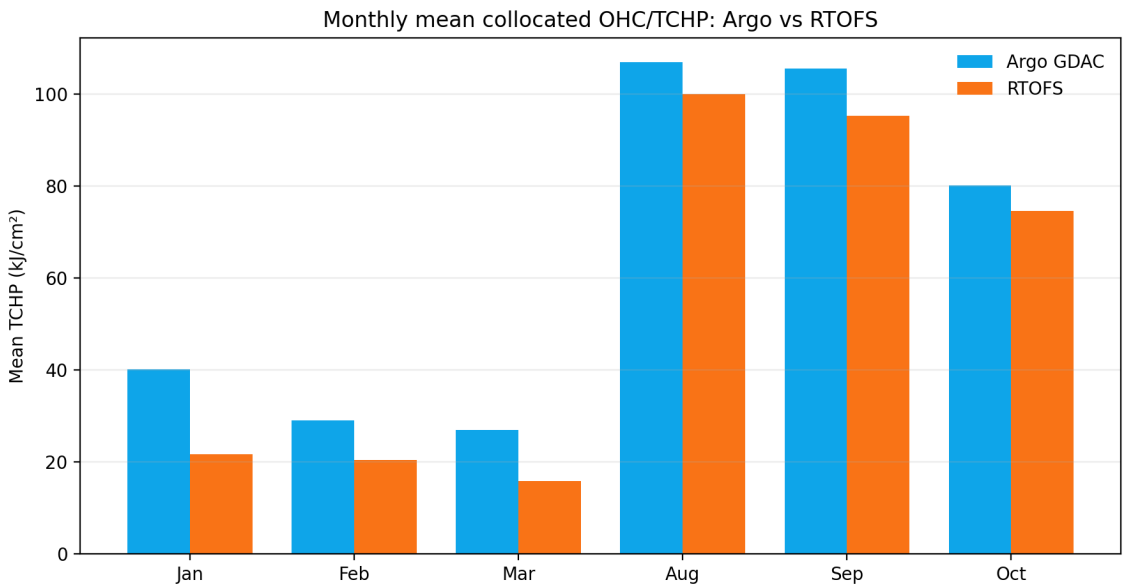

Figure 1: Monthly mean collocated TCHP for Argo GDAC and RTOFS.

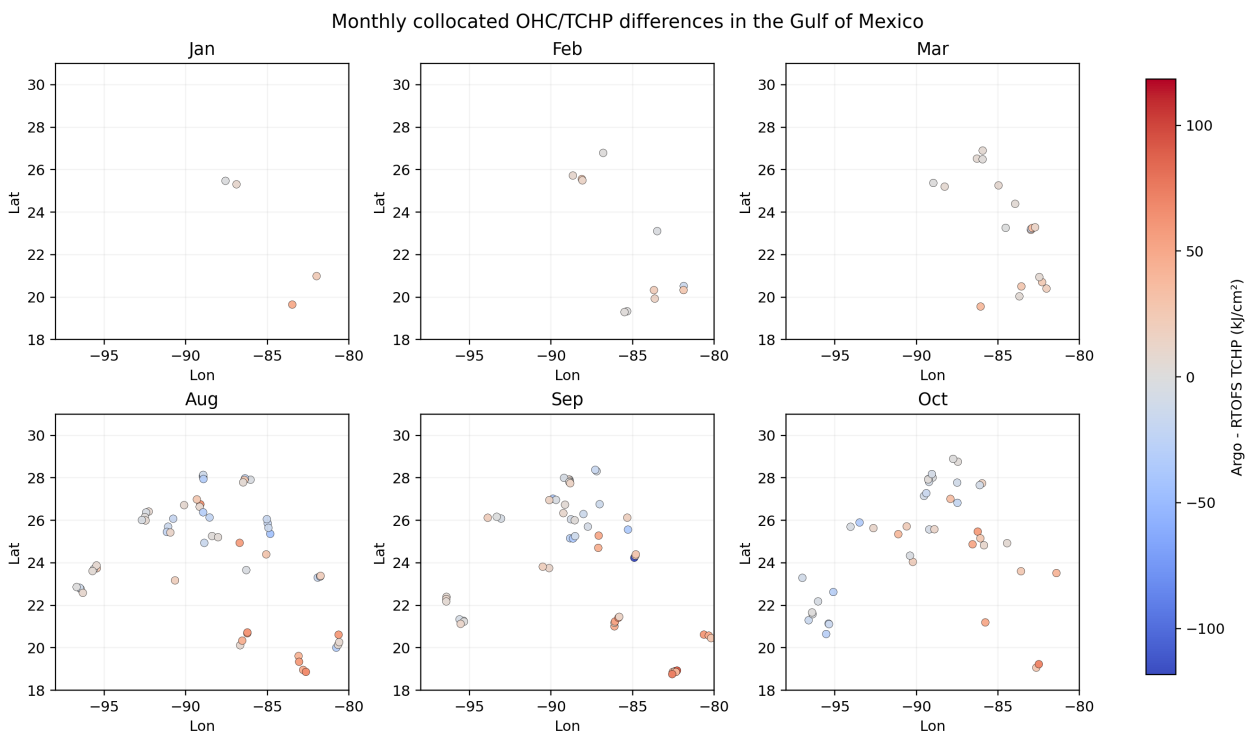

Figure 2: Spatial distribution of collocated monthly differences, computed as Argo minus RTOFS TCHP.

Interpretation

These monthly means are based on collocated profile dates rather than full-month daily model fields. That makes the comparison like-for-like at the same locations and times, which is the right framing for a model-vs-observation OHC benchmark.